

Use Case “View Product Details”

1. Use case code

UC003

2. Brief Description

This use case describes the interaction between Customer and AIMS when the customer wants to view the detailed information of a selected product (Book / CD / DVD / Newspaper) and may choose a quantity and add the product to the cart.

3. Actors

Customer

4. Preconditions

The customer has navigated to the product list/search results and selected a product. No login is required.

5. Basic Flow of Events (Narrative)

1. The customer selects a product from the list/search results to view its details.
2. The system checks whether the selected product exists and is available to be shown.
3. If the product exists, the system determines the product type.
4. Based on the identified type, the system retrieves the type-specific details together with the common information of the product.
5. The system displays the detailed information of the selected product.
6. The customer reviews the details and selects a corresponding quantity.
7. If the selected quantity is zero, the use case ends.
8. If the selected quantity is greater than zero, the customer chooses Add to Cart and confirms the action.
9. When confirmed, the system adds the product to the cart with the selected quantity and the use case ends.

10. If the customer cancels the add action, the use case ends with no change to the cart.

11. If, at Step 2, the product does not exist, the system shows “Product not Found” and the use case ends.

6. Alternative Flows

Table 1 — Alternative flows of events for UC “View Detail Product”

No	Location	Condition	Action	Resume location
1	At Step 2	productId not found	Software shows “Product not found”	End
2	At Step 6	Selected quantity equals 0	No item is added to cart	End
3	At Step 8	Customer cancels the add-to-cart action	No item is added to cart	End

7. Input data

None

8. Output data

Table A – Common information of selected product

No	Data fields	Description	Display Format	Example
1	Title	Name of product	String	Introduction to Artificial Intelligence
2	Price	Price of product	Non-negative integer	150000

3	Category	Category of product	String	book
4	Avail	Number of products available in store	Non-negative integer	12
5	Image	Image of product	Image (.png / .jpg)	book.png

Table B – Information of Book

No	Data fields	Description	Display Format	Example
1	Author	Author of the book	String	Andrew Ng
2	Publisher	Publisher of the book	String	MIT Press
3	Cover Type	Type of the cover	String	HardCover
4	Number of Page	Number of pages	Positive integer	420
5	Publication Date	Date when the book is published	Date	2025-02-27
6	Language	Language of book content	String	English
7	Genre	Genre type of book	String	Computer Science

Table C – Information of DVDs

No	Data fields	Description	Display Format	Example
1	Disc Type	Type of DVD	String	Blu-Ray
2	Director	Director of the movie	String	Christopher Nolan
3	Studio	Studio producing the movie	String	Warner Bross
4	Language	Primary spoken language in the movie	String	English
5	Subtitle	Available subtitle languages	String	Vietnamese, Japanese
6	Release Date	Date when the DVD was released	Date	2024-10-20
7	Genre	Genre of the movie	String	Science Fiction

Table D – Information of CDs

No	Data fields	Description	Display Format	Example
1	Artist	Name of artist or band	String	Taylor Swift”
2	Record Label	Music record label	String	Republic Record
3	Genre	Music genre	String	Pop

4	Release Date	Date when the album was released	Date	2022-11-18
5	Track list	List of songs in the CD (each includes title and duration)	String	1. Lavender Haze – 3:22; 2. Anti-Hero – 3:20; 3. Karma – 3:25

Table E – Information of Newspapers

No	Data fields	Description	Display Format	Example
1	Editor-in-chief	Chief editor responsible for publication	String	Pham Thanh Tung
2	Publisher	Publisher or press agency	String	Vietnam News Agency
3	Publication Date	Date of publication	Date	2025-03-05
4	Issue Number	Newspaper issue number	Non-negative integer	315
5	Publication Frequency	Frequency of release	String	Daily
6	ISSN	International Standard Serial Number	String	2616-0455
7	Language	Publication language	String	Vietnamese
8	Sections	Newspaper sections	String	Politics; Business; Technology; Culture; Sports

9. Postconditions

- The system has displayed the detailed information of the selected product.
- If the customer confirmed adding with a positive quantity, the product has been added to the cart with that quantity; otherwise, the cart remained unchanged.